STAFF REVIEW · H446 283 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Router.

2. A connection between two vertices.

3. Eight.

4. No. A router (or routing-capable layer-3 switch) is needed to route between IP networks.

I Do — source worked steps

1. Name resolution — The resolver returns the IP address for library.example (or uses a cached answer).

2. Transport connection — TCP establishes a connection to the server, typically destination port 443.

3. Application exchange — TLS secures the connection; HTTPS carries the page request and response.

We Do — guided CHECK

1. Identify the other layers
TCP provides reliable transport for SMTP; IP handles inter-network addressing and routing; link delivers frames on each local link.

You Do — source mark schemes

1. [2] Application layer [1]. DNS resolves domain names to IP addresses (among other DNS record types) [1].

2. [6] HTTP: application layer [1], handles web requests/responses [1]; TCP: transport layer [1], delivers an ordered reliable byte stream [1]; IP: internet layer [1], provides packet addressing and routing across networks [1].

3. [4] DNS performs name resolution, not website-content encryption [1]; TLS provides encryption for HTTPS traffic [1]; TCP is a transport protocol for reliable ordered data [1]; wireless channel selection is a Wi-Fi/link-layer concern [1].

Exit Ticket — source answer

1. True, though they perform different functions.